

Temporal Liquidity Crossing for Blockchains

An EIP-1559 Extension for Admission Below the Base Fee

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Abstract. Transactions differ in how their value changes with execution time and relative position. Ethereum’s EIP-1559 [2] lets them compete through fee fields, but a fee bid gives no enforceable claim to an earlier position in the block, and no way to express willingness to accept a later one. Its base fee is a hard admission floor. We introduce **temporal liquidity markets** (TLM), a framework for crossing complementary temporal requirements: transactions seeking early-region execution (consumers) can fund transactions that opt into exclusion from that region (providers). Selected consumers complete within the early region and are ordered among themselves by authorization density, both validator-checked. In our block-local EIP-1559 extension, providers may be admitted below the base-fee floor. Under TLM regulation, funding-conditioned and capacity-capped provider gas is excluded from the ordinary EIP-1559 controller input. Counting it there instead raises the base fee and displaces ordinary demand at a positive base-fee fixed point. Exclusion lets the controller regulate ordinary demand at its original target and preserves the no-TLM ordinary allocation by construction; in synthetic experiments it raises total utilization with funded gas. Selected consumers pay their signed authorizations; these payments fund provider shortfalls and the surplus is burned, so no subsidy enters. No inserted consumer bid can raise another consumer’s charge, and a builder that inserts its own provider loses on it.

1 Introduction

An arbitrage expiring in one slot and a treasury rebalance due this week can use identical EIP-1559 fields while having different losses from delay. Price can correlate with urgency, but the transaction format does not identify urgency, a deadline, or willingness to accept later execution.

Time is nevertheless fundamental to execution. Transactions arrive into finite service intervals, contend for ordered state transitions, and may lose value before capacity becomes available. Current blockchains represent this information indirectly: through arrival, tips, application-level deadlines, or the user’s choice of chain. Gas describes what execution consumes and fee fields describe payment limits; the temporal profile remains a secondary characteristic bundled into price and chain architecture.

1.1 Observed workload separation

Three broad application profiles stand out. **Latency-sensitive trading** requires rapid placement, cancellation, and relative ordering; Solana and Hyperliquid are prominent venues for this activity [13,16]. **Cost-sensitive payments and transfers** value low unit cost and can often tolerate delay; they are prominent on Tron and in the L2 ecosystem. **High-value settlement and capital-intensive DeFi** remain central to Ethereum mainnet.

The activity measures show the separation. In an August 2026 snapshot, Ethereum held \$49.76B of DeFi TVL, 8.4 times Solana’s \$5.90B, while Solana recorded \$3.46B of daily DEX volume, 1.84 times Ethereum’s \$1.88B [13,14]. Tron had 3.79M active addresses, 6.7 times Ethereum’s 569K, but little daily DEX volume [13,14]. The temporal requirements behind these profiles differ sharply. This paper asks whether a transaction can express some of that difference directly, rather than only through fees, latency investment, or its choice of chain.

1.2 From heterogeneous demand to mechanism scope

Applications can have different utility functions over execution time and position. Trading orders, liquidations, settlements, payments, oracle updates, and treasury operations can differ in sensitivity to delay, relative position, cadence, and expiry. A settlement transfer and an order-book action should not be modeled as homogeneous merely because both consume blockspace. Tokenized-asset settlement and fully on-chain order books illustrate these different settings [15,16].

We introduce TEP and TSP as components of a framework for building **temporal liquidity markets** (TLM). A **Temporal Execution Profile** (TEP) describes one request’s admissible positions, value over position, resource requirement, and expiry. A **Temporal Stream Profile** (TSP) describes cadence and continuity across blocks. They identify when a request seeking earlier service can fund a compatible request willing to execute later.

This framework can be studied on L1s and L2 sequencers. We use Ethereum because EIP-1559 supplies a mature baseline. The paper analyzes one opt-in, block-local TEP mechanism. It asks whether transactions that supply temporal liquidity can be funded and admitted without reducing the allocation of opt-out ordinary transactions, while selected temporal-liquidity consumers receive an assurance of completion within a stated early region of the block. For request i , let $q \in A_i$ denote a feasible execution position and define its TEP as

$$\theta_i = (A_i, V_i(q), r_i(q), d_i).$$

Here A_i is the admissible position set, $V_i(q)$ the value of execution at position q , $r_i(q)$ a state-dependent resource requirement, and d_i an optional expiry. The TEP is an analytical description; this paper does not propose encoding all of its fields in an Ethereum transaction.

Definition 1 (temporal liquidity). Temporal liquidity is the economically relevant capacity to move execution across admissible positions without violating feasibility, together with the value gained or surrendered by that movement.

1.3 A two-sided temporal-liquidity crossing framework

This subsection isolates complementary requirements for earlier and later execution before specifying EIP-1559 settlement. Consider two feasible positions q_E and q_L , with q_E earlier than q_L , and define transaction k 's value of moving earlier as

$$\Delta_k = V_k(q_E) - V_k(q_L) \geq 0.$$

Let urgent transaction u initially occupy q_L , and flexible transaction f initially occupy q_E . We call them a *temporal-liquidity consumer* (consumer) and a *temporal-liquidity provider* (provider), respectively. Let W_{before} and W_{after} denote their combined temporal value before and after their positions are exchanged:

$$W_{\text{before}} = V_f(q_E) + V_u(q_L), \quad W_{\text{after}} = V_f(q_L) + V_u(q_E).$$

The swap changes combined temporal value by $\Delta_u - \Delta_f$. The consumer's maximum temporal bid is Δ_u ; the provider's minimum temporal offer is Δ_f . A mutually acceptable crossing exists when $\Delta_u \geq \Delta_f$; strict inequality leaves positive temporal surplus.

We apply the temporal-liquidity crossing framework to Ethereum through a *Temporal Crossing Mechanism for EIP-1559*. Transactions submit bids or offers according to their temporal-liquidity roles: a consumer signs a positive temporal authorization as its bid for earlier service, while a provider's below-floor fee fields determine a shortfall, the minimum monetary offer required for accepting later service. In this framework, these correspond to the consumer's Δ_u and provider's Δ_f . Settlement uses a pool rather than a direct transfer: selected consumers pay their bids into it, the pool covers funded provider offers, and what it does not spend is burned. Here, "earlier" means completion within a defined early region of the block, while "later" means execution outside that region. The formal rule ranks qualifying consumers by authorization per unit of realized gas while retaining builder discretion over opt-out transactions and all ordering not expressly constrained.

A full assessment additionally models block capacity, state-dependent execution costs, and effects on other transactions and protocol stakeholders.

A request set is **temporally degenerate** over a horizon when moving any request among its admissible positions changes neither its value nor what can be executed. A TEP then adds no allocative information beyond a single-resource capacity model: admission reduces to

$$\sum_i x_i w_i \leq K_{\text{tot}}, \quad x_i \in \{0, 1\},$$

where w_i is request i 's fixed resource quantity, x_i its admission indicator, and K_{tot} the aggregate capacity for the horizon, both in the same unit: gas in the EIP-1559 specialization. Temporal mechanisms matter when time or position changes value or feasible execution.

1.4 Contributions

We make three contributions. (i) We introduce *temporal liquidity* as an execution-time dimension orthogonal to price and execution value, and use a minimal temporal execution profile to describe it. An ordinary fee bid supplies no offer to accept later execution and no enforceable claim to position. The mechanism below supplies both. (ii) We give a two-sided temporal-liquidity crossing framework and instantiate it as a block-local extension to EIP-1559. Selected consumers' conditional authorizations fund providers that opt into execution outside the early region, enabling admission of demand otherwise suppressed by the base-fee floor. Selected consumers receive validator-checked completion within that region and relative rank among selected consumers by authorization density. The crossing is internal and self-funded: consumer payments cover provider shortfalls, the surplus is burned, and no subsidy is required from the protocol or opt-out users. (iii) We analyze TLM-adjusted base-fee metering: gas of funded providers is excluded from the controller's metered input, so temporal-liquidity information informs base-fee control. At a positive base-fee fixed point, this preserves EIP-1559's ordinary-gas target and permits total block utilization above its 50% target. A reference-preserving admission rule adds providers only within residual capacity, retaining every transaction in the no-TLM reference block by construction rather than by measurement. We derive realized-settlement and provider-gas constraints, verify direct self-supply safety, and bound what a builder can gain from an inserted consumer bid.

2 Why temporal-liquidity crossing?

2.1 Limits of one-sided fee bidding

With base fee b , fee cap m , and priority-fee cap $p^{\max}$, EIP-1559 admits an ordinary transaction only if $m \geq b$; its effective tip is $\min\{p^{\max}, m - b\}$. Ethereum's consensus rules impose no intra-block ordering: descending priority-fee order is a client default rather than a protocol rule, and competitive builder auctions need not follow it [24]. A higher tip therefore raises inclusion probability and tends to place a transaction earlier by convention, but purchases no verifiable claim to any region of the block. A transaction with $m < b$ has neither an ordinary execution path nor a payment counterparty through which it can fund that execution. Lowering a priority-fee cap does not create an offer to execute later, while a consumer's higher payment cannot fund below-floor admission of a provider. A block-local crossing adds these missing links alongside EIP-1559's inter-block base-fee feedback.

2.2 Block-local temporal crossing

A consumer's positive authorization is conditional: nothing is charged unless the consumer is selected, and a consumer is selected only against funded providers. Selected consumers receive completion within a defined early region; the pool

covers providers' fee shortfalls in return for later execution. Validators check early-region completion and, among selected consumers, descending authorization per unit of realized gas. Opt-out transactions may still appear in the early region at builder discretion; all ordering not expressly constrained remains unchanged.

2.3 Related work

Babaioff and Nisan analyze EIP-1559 welfare with patient bidders [1]. Their patience is cross-block waiting without a payment transfer; our providers are compensated for later in-block execution and may be admitted below the base-fee floor. Kiayias et al. propose tiered price-delay schedules for heterogeneous urgency [17]; our two-sided crossing is not a menu of service tiers or tier-level price discrimination. Work on latency races, multidimensional fees, fee-market dynamics, empirical outcomes, patient users, and delay cost provides complementary context [3,4,6,7,8,9,11].

Mini-block auctions [18] and private-transaction evidence [19] support the premise that intra-block position can matter. Timeboost [12,20] exposes priority in production, while its concentration and reversion evidence leaves access and fairness open. Preconfirmations [21] provide advance, rather than block-local, assurance.

Roughgarden [10] evaluates transaction fee mechanisms by user incentive compatibility (UIC), myopic miner incentive compatibility (MMIC), and resistance to off-chain agreements. The miner there is the block producer, which in this paper is the builder. A first-price auction is MMIC and not UIC; second-price-type auctions fail MMIC, because the producer gains from bids it inserts. Chung and Shi [25] show that a mechanism that is UIC and proof against a producer colluding with one user must pay the producer nothing. Our consumer charge is first-price; Section 5 states what holds for inserted transactions and what is not claimed.

3 A two-sided mechanism for EIP-1559

Let b be the block base fee. For transaction i , the retained EIP-1559 fields are `max_fee` = m_i , `max_priority_fee` = $p_i^{\max}$, gas limit L_i , and realized charged gas $g_i \leq L_i$. It additionally signs one lump-sum **Temporal Liquidity Authorization** (TLA) denominated in wei. We call the signed value the transaction's *declaration*; a consumer's declaration is also called its authorization or its bid. A transaction with $\text{TLA} = 0$ is neutral (opt-out) and follows ordinary EIP-1559 treatment. Block positions are measured by cumulative charged gas; the *early region* is the prefix ending at E .

3.1 Consumers

A consumer signs $\text{TLA} = c_i > 0$. It is a lump-sum temporal authorization, distinct from the gas-priced EIP-1559 fee fields. Let g_i be its realized charged gas in the executed block and define its authorization density as

$$\rho_i = \frac{c_i}{g_i}.$$

The mechanism fixes a protected consumer-selection gas allowance $K_C = \kappa_C K$, where K_C has gas units, K is the block gas limit, and $\kappa_C \in [0, 1]$ is the dimensionless allowance fraction. K_C caps the realized gas of consumers selected for protected service; it is not the size of the early region. Protected service is sold only against provider supply: the consumer allowance of a block is

$$A_C = \min\{K_C, \lambda G_P\},$$

where G_P is the realized gas of the funded providers (§ 3.4) and $\lambda > 0$ is a protocol parameter. We call this dependence the *allowance link*; it binds when $\lambda G_P < K_C$. A candidate is eligible if it satisfies ordinary EIP-1559 validity, including $m_i \geq b$, appears in the proposed block, and meets the *consumer floor* $\rho_i \geq \beta b$, where $\beta > 0$ is a protocol parameter. Positive TLA does not guarantee inclusion or protected service. Under the stated selection policy, the builder processes eligible candidates by decreasing ρ_i , with transaction hash breaking ties, and adds a candidate to the selected set C^* if its realized gas fits the remaining A_C ; otherwise it skips the candidate. Density and the allowance use realized gas, as the provider side does (§ 3.4). The signed gas limit remains an EIP-1559 validity and liability bound and plays no part in selection, so a transaction takes allowance only by the gas it uses, on which it also pays the base fee. The allowance makes protected service scarce, so a large collection of low-density positive-TLA bids cannot all qualify. Only selected, included consumers belong to C^* and are charged. Excluded and unselected positive-TLA transactions pay no temporal charge and remain ordinary transactions if included.

A selected consumer pays its ordinary EIP-1559 charge plus a temporal charge equal to its authorization, $a_i = c_i$ (§ 3.4). At validity checking, the sender must back its ordinary EIP-1559 maximum liability, transferred value, and c_i . A consumer therefore knows its temporal charge when it signs: c_i if it is selected, and nothing otherwise. Consumer selection is a constrained multi-unit allocation problem; this paper uses the stated deterministic greedy rule rather than the ascending-bid auction of Ausubel [22].

3.2 Consumer position assurance

TLA does not prescribe an exact transaction order. The initial mechanism uses a fixed early completion region

$$E = \chi K$$

with $\chi = 0.20$ and $K_C = 0.10K$ in the initial parameterization. “Early” is therefore a logical position, not an arrival or wall-clock time within the slot. For the final execution order $\preceq$, let

$$h_i = \sum_{k: k \preceq i} g_k$$

be the cumulative charged-gas endpoint of transaction i ; its start position is $h_i^- = h_i - g_i$. A valid block requires $h_i \leq E$ for every $i \in C^*$. Thus a selected consumer is fully executed before the early-region threshold is crossed. The two quantities answer different questions: K_C determines which scarce set of consumers qualifies for protected service, while E determines where that selected set must finish. The early region is not reserved solely for consumers: at the builder's discretion, a neutral transaction may appear before or among selected consumers, subject to transaction validity and state dependencies.

3.3 Providers

A provider signs $TLA < 0$, chooses m_i with $\nu b \leq m_i < b$, and accepts execution outside the early region, after completion of the protected consumer set. For an effective provider tip p_i , its per-gas shortfall is

$$s_i = b + p_i - m_i.$$

The builder selects the effective tip subject to

$$0 \leq p_i \leq \min\{p_i^{\max}, (1 - \eta) m_i\}, \quad 0 < \eta < 1.$$

The provider branch retains the signed relation $p_i^{\max} \leq m_i$ and adds two rules. The *fee-cap floor* makes a provider eligible only if $\nu b \leq m_i < b$, with $0 < \nu < 1$. It is checked against the base fee of the including block, like the consumer floor, and it decides eligibility, not payment: a provider never pays more than $g_i m_i$. The *tip cap* is the second term of the bound above. It is a clamp, as the ordinary EIP-1559 effective tip is: a provider that signs a larger $p_i^{\max}$ remains eligible, and the builder takes less than it authorized. It is checked on the published effective tip and two signed fields, and does not depend on the base fee of the including block. Both η and ν are protocol parameters. Together they give $m_i - p_i \geq \eta m_i \geq \eta \nu b$ and $s_i \leq b - \eta m_i$: at least a share η of what a provider pays goes to the base fee, and the pool never pays the whole base fee for a provider. Both are bounds and are not fitted to traffic. With $\eta = 0$ only the EIP-1559 relation $p_i^{\max} \leq m_i$ remains: a provider may pay its whole fee cap as tip, the pool then pays the whole base fee, and a builder's own provider costs the builder nothing. A small positive η closes that case and stays close to the tip a builder may take today. The builder may take the full capped tip, or publish a lower p_i when that maximizes its block revenue. The selected p_i is published in consensus-visible block data, and validators recompute shortfall, settlement, and builder revenue from it. For realized gas g_i , the provider pays $g_i m_i$; the pool pays $g_i s_i$; the protocol burns $g_i b$; and the builder receives $g_i p_i$. Thus:

$$g_i m_i + g_i s_i = g_i b + g_i p_i.$$

The base fee remains fully paid and burned. What changes is the incidence: the provider's own payment may be below b because consumers cover the difference.

3.4 Provider selection, execution, and settlement

The builder executes a candidate final block before proposing it. Provider funding is not an entitlement: together with consumer selection (below), the builder chooses a funded-provider set $\mathcal{F}$ from eligible negative-TLA candidates (transactions with signed negative TLA, $\nu b \leq m_i < b$, valid retained EIP-1559 fields, and a feasible placement outside E), subject to final funding, provider-gas, and early-region constraints. The protocol checks feasibility of the final set rather than a globally observable optimum. A provider below the base fee is in a block only if it is funded, so a block holds no record of candidates that were passed over, and a priority rule among providers could not be checked without a protocol-visible list of candidates. For reproducibility, our simulator scans eligible providers by increasing shortfall, with transaction identifier ties.

Let g_i be the realized gas of provider $i \in \mathcal{F}$, and define

$$S_i = g_i s_i, \quad S = \sum_{j \in \mathcal{F}} S_j.$$

Thus S_i is provider i 's funded amount and S is the endogenous total provider funding. The block is funding-valid only if $S \leq R$, where $R = \sum_{i \in C^*} c_i$ is the total payment of selected consumers. Each block t has a provider-gas allowance $\bar{P}_t$, a protocol parameter that caps total realized gas of funded providers: $\sum_{j \in \mathcal{F}} g_j \leq \bar{P}_t$. The final block must also obey Ethereum's hard gas limit. A provider's gas limit remains its normal signed EIP-1559 bound and sender-liability bound; it does not reserve TLM funding. The proposed block identifies C^* and each effective provider tip p_i . On replay, validators check signed declarations, the fee-cap floor, the tip cap, the consumer floor, consumer selection under A_C , consumer density order and early completion, provider placement after E , shortfalls, settlement, the provider-gas allowance, and hard gas validity. Verification is relative to the proposed block: validators confirm that the included set obeys the rules, not that it is the set a builder should have chosen, since transactions absent from a block leave no record and builders do not share a mempool.

Joint selection. The selected consumers C^* and the funded providers $\mathcal{F}$ constrain each other: the allowance $A_C = \min\{K_C, \lambda G_P\}$ grows with funded provider gas $G_P = \sum_{j \in \mathcal{F}} g_j$, and providers can be funded only up to the consumers' total payment R . Neither set can be fixed before the other. Validators do not search for the pair $(C^*, \mathcal{F})$. They check that the pair in the proposed block is consistent: C^* is what the selection rule of § 3.1 yields under A_C , and $\mathcal{F}$ satisfies the funding and provider-gas constraints with that R . A builder can find a consistent pair by starting from all candidates and only removing: select consumers under the current allowance, compute R , drop providers that R cannot fund, recompute A_C , and repeat, never restoring a candidate it has removed. With realized gas held fixed, each pass can only lower the allowance, so the procedure ends; its extension to realized gas that changes with the order is left to future work (§ 7). A consistent pair always exists: with no funded provider, $G_P = 0$, no consumer is selected or charged, and the block is an ordinary EIP-1559 block.

Settlement is pay-as-bid. Every selected consumer pays its signed authorization, and providers are funded from the total:

$$a_i = c_i \quad (i \in C^*), \quad R = \sum_{i \in C^*} a_i, \quad S = \sum_{j \in \mathcal{F}} S_j \leq R.$$

The surplus $R - S$ is burned. No balance is carried between blocks and no external funding enters, so

$$\sum_{i \in C^*} a_i = R = S + (R - S).$$

Burning is the destination that no participant can turn to its own account within the block. Returned to consumers in proportion to their authorizations, the surplus reproduces the pro-rata rule set aside in § 3.6. Paid to the builder, it would reward higher bids. Paid to providers beyond their shortfalls, it would reward transactions that exist to collect it. Carrying the surplus to later blocks as a reserve is a multi-block design outside this paper. Pro-rata cost sharing has a related literature, including robust double auctions and excludable-public-good mechanisms [5,23].

3.5 Ordering

Provider selection and logical execution order are distinct. The builder may use shortfall, expected revenue, or state dependencies when choosing providers; the protocol checks only the final set's realized constraints. To check consumer order, delete every transaction other than those in C^* from the final block order. The remaining list must be sorted by nonincreasing authorization density ρ_i , with transaction hash breaking ties. Every funded provider j must begin outside the early region, $h_j^- \geq E$. Neutral transactions may appear before or among consumers, provided $h_i \leq E$ for all $i \in C^*$. The mechanism constrains three things: completion of C^* within E , the density order within C^* , and the start position of funded providers. Block composition, provider order, and all else remain at builder discretion, as under EIP-1559 and proposer-builder separation (PBS) today.

3.6 Declaration consequences

A positive declaration c_i is a sender-backed liability and, through $\rho_i = c_i/g_i$, the input to consumer selection and to relative order within the consumer subsequence. If the consumer is selected, its temporal charge is its declaration, $a_i = c_i$. Declaring more improves selection rank and position among selected consumers, and costs the amount declared. A declaration below the consumer floor $\beta b g_i$ does not qualify, and when consumer demand exceeds the allowance, low-density candidates are skipped.

The rule is first-price. A consumer's charge is set by its own declaration and by no other transaction. It is therefore not UIC in the sense of [10]: a consumer gains by declaring less than its value, as far as it can and still be

selected, and its best declaration depends on declarations it does not see. Where declarations are visible before the block, rivals can also outbid one another in steps [26]. A declaration above value does not pay, since it is charged in full. EIP-1559’s priority fee is also paid as bid, but it buys no enforceable position, and under EIP-1559 the posted base fee does the rationing when blocks are not full [10]; the temporal charge has no posted price beyond the consumer floor. Section 6 reports how funded provider gas varies with the level of declarations.

Two alternatives were set aside. (i) *Pro rata*. Charging $a_i = c_i S / R$, with $R = \sum_{i \in C^*} c_i$ the total authorized, and refunding $R - S$ is budget-balanced, but it bounds every charge by the funded shortfall S , an amount that providers determine. A consumer that values rank above S gains by inflating its declaration, limited only by the balance it can back, and when $S = 0$ protected service is free. (ii) *Second price*. Charging each selected consumer the next density in the selection order removes that incentive, but lets one transaction’s declaration set another’s charge. A builder earns the tips of the providers it funds, so it gains by inserting a bid just below a selected consumer (§ 5); second-price rules are the standard example of this failure [10]. In our traffic model the second-price rule also left 94% of selected consumers able to gain by declaring less, and it funded the same provider gas as pay-as-bid once each consumer paid the least at which it was still selected (§ 6).

A negative declaration opts the transaction into the provider branch: it executes outside E , and funded admission requires a fee cap below the base fee. The provider pays $g_i m_i$; the pool pays the shortfall needed for the burn and builder tip.

Lemma 1 (negative-TLA magnitude is inactive). *Holding the fee fields, realized gas, and other block inputs fixed, a provider’s negative TLA value affects neither its shortfall, settlement, funding feasibility, nor ordering requirement.*

Proof. Only the sign selects the provider branch; shortfall and settlement are defined from the fee fields and realized gas, and placement is after E .

The mechanism is block-local. It reallocates position and payment inside one block; it does not defer a transaction across blocks or create physical capacity.

4 Evaluating temporal liquidity in a blockchain economy

4.1 Evaluation framework

The temporal-liquidity framework models a blockchain economy as a mix of applications whose demand differs in urgency, delay tolerance, expiry, and position value. Traffic mix is therefore an economic input, not merely a simulation detail: it determines whether complementary temporal demand and supply can cross. Capacity alone is not a sufficient benchmark, because more admission can displace other demand or change incidence. We report settlement, admission and displacement, controller response, and incidence. The reference-preserving admission rule, defined in § 4.4, retains the no-TLM reference block; consumer charges are the signed authorizations of selected consumers.

The synthetic traffic model has positive-TLA consumers, below-floor providers, and neutral transactions. Consumers seek early-region service and sign heterogeneous authorizations, so that selection and rank vary with willingness to pay for earlier execution; providers accept execution outside that region; neutral transactions retain the baseline path. These are economic roles, rather than fixed application labels. Homogeneous traffic or traffic without providers reduces to the EIP-1559 baseline. Let their offered-gas shares be

$$(\pi_C, \pi_P, \pi_N), \quad \pi_C + \pi_P + \pi_N = 1.$$

Our simulator compares baseline EIP-1559 with two TLM variants. The main offered-gas mix is $(\pi_C, \pi_P, \pi_N) = (1/5, 1/3, 7/15)$. Additional sensitivity runs set $\pi_P = 1/3$, vary $\pi_C \in \{1/10, 1/5, 1/3\}$, and assign the residual share to π_N . In the $\pi_C = 1/3$ run, the offered gas of positive-TLA consumer candidates exceeds the protected consumer-selection allowance $K_C = 0.10K$, so the allowance is active and candidates are skipped. Runs contain 600 blocks (200 warm-up) over 20 fixed seeds. The hard limit is $K = 60$ million gas, the limit Ethereum adopted in November 2025, and the EIP-1559 target is $g^* = 30$ million gas. Provider-gas allowances are $\bar{P}_t \in \{0.20K, 0.30K\}$, and the early-region capacity is $E = 0.20K$; each is measured in gas. The consumer floor is $\beta = 0.10$, the link ratio is $\lambda = 1$, the tip cap is $\eta = 1/4$, and the fee-cap floor is $\nu = 1/2$: the builder may take up to three quarters of a provider's fee cap, and a builder's own provider costs it at least $b/8$ per gas. Neither provider rule binds for the simulated fee fields. The simulated builder takes the full authorized tip for every funded provider; a builder optimizing over p_i could fund more providers at a lower tip, so the reported builder revenue is a lower bound under this policy. Ordinary and consumer fee caps follow a capped Pareto distribution with tail exponent 1.35, and consumer authorizations are drawn as realized gas times a uniform factor on $[4, 12]$. Declarations are thus drawn, not chosen: the simulation does not model bidding, and Section 6 reports sensitivity to their level. Because simulated gas is independent of fee caps, this exponent is also the base-fee elasticity of ordinary gas demand; other inputs are synthetic. Results report 95% Student- t intervals. The simulator checks selection, settlement, early-region completion, admission, and the gas measure used by the base-fee controller.

4.2 TLM regulation for the base fee

Let O_t denote controller load from neutral and consumer transactions (not only TLA-zero transactions), and let P_t denote funded provider gas. We compare *naïve metering* with *TLM regulation*. Under naïve metering, the EIP-1559 controller measures $G_t = O_t + P_t$, so provider gas is included in the controller load. Under TLM regulation, the controller measures $G_t = O_t$. Both use the EIP-1559 base-fee feedback rule [2,6,10].

$$b_{t+1} = b_t \left[1 + \gamma \frac{G_t - T}{T} \right],$$

where $T = g^*$ is the EIP-1559 target and $\gamma > 0$ is the adjustment parameter. A base-fee fixed point means $b_{t+1} = b_t > 0$: the controller leaves the base fee unchanged from one block to the next. Provider gas is excluded because it is not admitted through the same ordinary EIP-1559 base-fee regulation as O_t : its admission is conditional on temporal-liquidity funding from consumers and bounded by the TLM provider allowance. The TLM controller therefore treats O_t as ordinary congestion and P_t as a separately funded, capacity-capped addition.

Proposition 1 (fixed-point identity). *Under naïve metering, if the base fee is at a positive fixed point, then $O_t + P_t = T$. Thus, funded provider gas uses part of the metered target that would otherwise be available to other ordinary transactions.*

Proof. At a fixed point, divide the update by b_t to obtain $1 = 1 + \gamma(G_t - T)/T$. Since $\gamma > 0$ and $T > 0$, $G_t = T$. Under naïve metering, $G_t = O_t + P_t$.

Proposition 1 is the negative half of the comparison: when all execution is included in the controller load, funded provider gas cannot raise fixed-point total utilization above T/K ; it replaces other metered gas. TLM regulation is designed to retain the ordinary controller target while admitting bounded, funded provider gas in addition.

In the main mixture, naïve metering leaves utilization at the baseline level, 50.07% against 50.05%: it admits 5.86 million provider gas per block, but ordinary admitted gas falls from 30.03 to 24.18 million and the mean base fee rises from 28.54 to 33.37. Metered gas stays at target while funded provider gas displaces ordinary admission. Under TLM regulation with 20% and 30% provider allowances, ordinary gas remains 30.03 million at a 28.54 base fee and total utilization reaches 68.49% and 69.31%.

4.3 Base-fee increase under naïve metering

Proposition 2. *Let $D(b)$ be ordinary gas demand at base fee b , b_0 the baseline fixed-point base fee with no funded provider gas, and b_1 the naïve-metering fixed-point fee with funded provider gas $P^* = \phi g^*$. Suppose $0 \leq \phi < 1$, $\epsilon > 0$ is the absolute base-fee elasticity of D , $D(b_1)/D(b_0) = (b_1/b_0)^{-\epsilon}$, baseline demand clears at the EIP-1559 target $D(b_0) = g^*$, and the hard limit does not bind. Then:*

$$\frac{b_1}{b_0} = (1 - \phi)^{-1/\epsilon}.$$

Proof. At the new fixed point, ordinary demand is $D(b_1) = g^* - P^* = g^*(1 - \phi)$. Constant-elasticity demand gives $D(b_1)/D(b_0) = (b_1/b_0)^{-\epsilon}$. Equating the expressions gives the result.

The point is to quantify the price of naïve metering: provider gas P^* forces ordinary gas down to $g^* - P^*$, so the base fee rises to ration ordinary demand. In the 20%-consumer run, $P^*/g^* = 0.195$; Proposition 2 predicts $b_1/b_0 = 1.175$ under the synthetic Pareto fee-cap exponent $\epsilon = 1.35$, versus 1.169 in simulation. The remaining 0.5% difference reflects the capped Pareto generator and indivisible transactions, whereas Proposition 2 assumes an uncapped constant-elasticity demand curve.

4.4 A bounded TLA-funded provider allowance

Consumer funding creates financial capacity, not physical gas capacity. Let K be the hard block gas limit and T the EIP-1559 gas target used by the ordinary congestion controller, with $T < K$; O_t denotes neutral and consumer gas, and P_t funded provider gas. Keeping the base fee unchanged does not itself preserve ordinary transactions: providers can still displace them within K .

Consumers are selected under the allowance rule of § 3.1, giving total consumer payment $R = \sum_{i \in C^*} c_i$ in wei. We then use a *reference-preserving admission rule* with two stages. Stage 1 constructs a reference ordinary block B_t^0 , the block that would have been built from neutral transactions and consumers without TLM, with realized gas H_t^0 ; our simulator packs it greedily by effective EIP-1559 priority fee, ties broken by a fixed identifier, so the same input always yields the same reference block. Stage 2 adds providers without removing any transaction in B_t^0 ; every consumer candidate is in B_t^0 whether or not it is selected, so the set of ordinary transactions does not depend on selection. The builder executes the final block; provider admission then has exactly two realized constraints:

$$\sum_{j \in \mathcal{F}_t} g_j s_j \leq R, \quad \sum_{j \in \mathcal{F}_t} g_j \leq A_t, \quad A_t = \min\{\bar{P}_t, K - H_t^0\}.$$

The provider-gas allowance $\bar{P}_t$ is defined in § 3.4; A_t is the residual provider-admission budget that additionally preserves the reference block. The first constraint says selected consumers fund actual provider shortfalls; K_C bounds this funding side by limiting who can be selected. The second caps actual provider gas and preserves the reference block. Both constraints are evaluated on realized gas, so no pool funding is committed before execution.

Lemma 2 (non-displacing provider admission). *Under the reference-preserving admission rule, every transaction in B_t^0 is retained and providers are added only within residual capacity. The completed block is therefore physically feasible, and realized gas increases by funded realized provider gas (equivalently, utilization increases by P_t/K).*

Proof. The first stage is retained. The realized provider-gas constraint ensures that provider gas fits within $K - H_t^0$. Since the ordinary allocation is unchanged, the increase in realized gas is funded provider gas.

Under TLM regulation, provider gas is excluded from the ordinary congestion signal, so the controller sees only O_t :

$$b_{t+1} = b_t \left[1 + \gamma \frac{O_t - T}{T} \right].$$

At a positive fixed point of the TLM controller, the same calculation as Proposition 1 gives $O^* = T$. Together with Lemma 2, any funded $P^* > 0$ gives

$$U^* = \frac{T + P^*}{K} > \frac{T}{K}.$$

For $T = K/2$, TLM regulation therefore raises total utilization above 50% while preserving the ordinary base-fee controller target and the reference ordinary

allocation. With the same ordinary demand curve and target, TLM regulation retains the baseline fixed-point base fee; providers remain subject to the hard block limit and provider allowance.

4.5 Heterogeneous preferences and Pareto comparison

Let $V_i(q_i)$ be request i 's gross value, in utility units, when it completes at logical block position q_i , and let z_i be its total monetary payment. We use the simple separable utility model

$$U_i(q_i, z_i) = V_i(q_i) - \mu_i z_i, \quad z_i \leq B_i.$$

Here $\mu_i > 0$ converts money into utility loss (payment sensitivity), and B_i is a budget. These are behavioral variables, not transaction fields. Let q_i^0 denote the consumer's position in the no-TLM reference block and q_i^1 its realized position under TLM. A selected consumer is individually rational only if its temporal value gain covers its realized TLA charge:

$$V_i(q_i^1) - V_i(q_i^0) \geq \mu_i a_i.$$

A provider has several outside options: raise its fee cap to $m_i \geq b$ and seek ordinary EIP-1559 admission, defer or cancel, or use another venue. Let U_i^{out} denote the best such utility. Funded provider participation is individually rational only if its TLM utility is at least U_i^{out} ; the pool payment covers the provider's shortfall and is not cash received by that provider. The reference-preserving rule makes the incidence by party explicit:

- **Reference-block neutral senders:** no funded provider displaces them.
- **Consumers:** selected consumers receive early-region completion assurance and relative rank among selected consumers by authorization density, and pay their signed authorization.
- **Providers:** funded providers can execute below the base-fee floor, outside the early region, if this dominates their outside option.
- **Builders:** for each funded provider j , the builder receives effective-tip revenue $g_j p_j$ without replacing a reference transaction's fee.
- **Protocol and ETH holders:** each funded provider still burns $g_j b$. Relative to the reference block, this adds ETH burn and reduces ETH supply, holding other effects fixed; the surplus of consumer payments over funded shortfalls is burned as well.

Two of these hold by construction rather than by measurement: retention follows from Lemma 2, and each side's payment is bounded by its own signed declaration, $a_i = c_i$ for consumers and m_i per gas for providers. Participation is then individually rational for any sender that authorizes no more than the movement is worth to it. Whenever both individual-rationality conditions hold, the rule is a weak Pareto improvement on the reference block: no user, builder, or ETH holder is worse off and capacity is respected. This includes neutral senders. A

neutral transaction keeps its fee and its admission, which is all that EIP-1559 promises it. It holds no claim to a region of the block: its place is at the builder's discretion with or without TLM, as it is today when a builder accepts private order flow, and it may still appear in the early region. Our simulator assigns no numerical V_i , μ_i , budgets, outside options, or builder operating costs; its results are allocation and accounting results.

5 Incentives: self-supply and inserted bids

Self-supply risk. MMIC requires that a builder cannot increase its current payoff merely by inserting a transaction it controls. This check is necessary because the consumer-funded pool pays the fee shortfall of funded providers: without a sender-side constraint, a builder could fabricate a provider transaction and divert consumer funding to itself.

The test is direct. Suppose the builder inserts a provider of its own. As the sender it pays that transaction's fee, $g_i m_i$; as the builder it collects the tip, $g_i p_i$, while the pool covers the transaction's shortfall. If the tip exceeds the fee, the insertion is profitable and MMIC fails. We call the requirement that it does not the *sender-backing condition*: on any provider, the builder receives no more than the sender pays. EIP-1559's burn already prevents a builder from recovering the base fee from its own transaction; a pool that pays part of a transaction's fee can reintroduce self-dealing, so the condition must be checked. The next proposition verifies it for the current mechanism.

Proposition 3 (direct self-supply safety). *The mechanism in § 3 satisfies the sender-backing condition for any builder-selected effective tip within the tip cap: a funded provider pays m_i per gas, the builder receives p_i , and builder net on a self-supplied provider is $g_i(p_i - m_i) \leq -\eta g_i m_i \leq -\eta \nu b g_i < 0$, by the tip cap $p_i \leq (1 - \eta)m_i$ and the fee-cap floor $m_i \geq \nu b$.*

The protection comes from the sender-paid fee cap, the tip cap, and the fee-cap floor; without the floor the loss could be made arbitrarily small. It holds for any effective tip the builder selects, and it addresses this direct self-supply strategy.

Inserted consumer bids. A builder earns the tips of the providers it funds, and consumer payments decide how many are funded. Under a rule in which one bid sets another's charge, that is a reason to insert bids. Under pay-as-bid it is not.

Proposition 4 (inserted consumer bids). *Let the builder insert a consumer transaction with declaration $\hat{c}$ and realized gas $\hat{g}$ into a block in which, without it, selected consumers pay R_0 , the funded providers receive S_0 , and $R_0 - S_0 \geq 0$ is burned. (a) With or without the insertion, every other consumer is charged its own declaration if selected and nothing otherwise. (b) Suppose the other selected consumers are unchanged and the builder funds an additional set D of providers. Its payoff changes by*

$$\Delta = \sum_{j \in D} g_j p_j - \hat{c} - b \hat{g} \leq (R_0 - S_0) - \sum_{j \in D} g_j (b - m_j) - b \hat{g}.$$

Hence the insertion loses whenever $R_0 = S_0$, and no insertion gains more than the payments that were about to be burned.

Proof. (a) holds by the settlement rule of § 3.4. For (b), the inserted transaction pays $\hat{c}$ as a consumer and burns $b\hat{g}$; its own tip returns to the builder. The funding constraint gives $\sum_{j \in D} g_j s_j \leq \hat{c} + (R_0 - S_0)$, and $g_j p_j = g_j s_j - g_j(b - m_j)$ with $m_j < b$.

Funded from its own payment, an inserted bid therefore cannot pay for itself: a provider's tip is always below its shortfall. It must also meet the consumer floor and fit the allowance like any other bid, and when the allowance is full it displaces a consumer whose payment then leaves the pool. The case that remains uses payments already made. If the residual $R_0 - S_0$ is close to the cost of one more provider, the builder can close the gap and collect that provider's tip, which is at most $(1 - \eta)g_j m_j$. If the inserted bid displaces no selected consumer, every consumer then pays what it would have paid and one more provider executes; the residual, which would have been burned, pays that provider's shortfall instead. The protocol cannot tell a builder-affiliated consumer from any other, and the bid meets the same conditions and pays the same charge either way. When the allowance link binds, added funding can also enlarge A_C and admit further consumers, each paying its own declaration for the service it requested; the bound then includes those payments. Section 6 finds that this is where the gain is large when tips are close to fee caps, and that only a tight tip cap limits it.

We do not claim MMIC in the sense of [10], which asks that a builder never gain from inserting transactions, whoever else is affected. We claim Propositions 3 and 4, which are direct self-supply properties. Section 6 measures how often the residual case arises. Resistance to agreements between a builder and users outside the protocol, the third property of [10], is left to future work: the burned surplus is where such an agreement would gain, since payments that would be burned could instead be shared between them.

Candidate-set selection is unconstrained here, as it is under EIP-1559 today. TLM adds a motive rather than a capability: protected service is scarce, so a builder can favour an affiliate by omitting a higher-density consumer. Under pay-as-bid the omission does not change what the affiliate pays, and Proposition 3 still binds, so the gain is placement rather than extraction.

6 Simulation study for TLM regulation under provider-gas bounds

This section evaluates TLM regulation with provider-gas allowances of 20% and 30% of the hard block limit. In both variants, funded provider gas is excluded from the ordinary EIP-1559 controller and separately bounded by the realized provider-gas allowance. Baseline EIP-1559 and naïve metering are comparison cases: naïve metering instead includes funded provider gas in the controller, which raises the base fee at a positive fixed point and reduces gas admitted through the ordinary channel.

The main mixture gives the following steady-state means. Parentheses contain 95% confidence half-widths across 20 seeds.

Table 1. **Baseline** is unmodified EIP-1559. **N** is naïve metering: it applies TLA while metering provider gas against the ordinary EIP-1559 target. **T20** and **T30** apply TLM regulation with a TLA-funded provider-gas allowance of, respectively, 20% and 30% of the hard block limit.

case	base fee	total utilization (%)	ordinary gas (million)	funded provider gas (million)	builder revenue (million)	builder revenue change
Baseline	28.54 (0.05)	50.05 (0.01)	30.03 (0.009)	0	49.89 (0.07)	baseline
N	33.37 (0.05)	50.07 (0.01)	24.18 (0.041)	5.86 (0.041)	46.71 (0.06)	−6.37%
T20	28.54 (0.05)	68.49 (0.08)	30.03 (0.009)	11.06 (0.045)	61.78 (0.08)	+23.84%
T30	28.54 (0.05)	69.31 (0.12)	30.03 (0.009)	11.55 (0.072)	62.34 (0.10)	+24.96%

The simulation implements the analytical contrast. T20 and T30 retain the baseline ordinary gas (30.03 million) and base fee (28.54), while raising total utilization from 50.05% to 68.49% and 69.31% through, respectively, 11.06 and 11.55 million funded provider gas per block. N instead remains at 50.07% utilization, raises the base fee 16.9%, and reduces ordinary gas 19.5%. Selected consumers pay 53.76 million per block. Under T20, 49.98 million of it funds providers and 3.77 million (7.0%) is burned; under T30 the burn is 0.47 million (0.9%). Two constraints therefore share the work in this mixture: consumer payments bound funding, and under T20 the provider-gas allowance binds in part of the blocks (realized provider gas averages 11.06 million against an allowance of 12.00 million, and 11.55 against 18.00 under T30). The allowance link rarely binds under T20 and T30 (0.1% of blocks), because funded provider gas exceeds K_C ; it binds in 45% of blocks under N. The preserved ordinary gas and base fee follow from Lemma 2 rather than from measurement; the simulation adds the magnitude of funded provider gas and the incidence. Funded shortfall never exceeds consumer payments, and no reference-block transaction is removed. Appendix A reports payment incidence and Appendix B reports consumer-load sensitivity.

Sensitivity to declarations. The simulation draws declarations, and under a first-price rule consumers have reason to declare less than they would pay (§ 3.6). Table 2 varies the level of declarations under T20. The last row charges each selected consumer the least at which it is still selected when the others keep their declarations; it uses every eighth block, on which the first row gives 11.05 million.

Table 2. T20 in the main mixture, by level of consumer declarations: consumer payments per block, funded provider gas per block (both in millions), and total utilization. Ordinary gas is 30.03 million and the base fee 28.54 in every row.

declarations	paid (M)	provider gas (M)	utilization (%)
declarations as drawn	53.76	11.06	68.49
every declaration 20% lower	43.01	9.97	66.67
every declaration 40% lower	32.00	8.32	63.91
least charge at which still selected	36.70	8.97	65.0

Total utilization is 68.5% at the drawn declarations and about 65% when consumers pay the least that keeps them selected. The structural results do not depend on the level of declarations; the utilization figure does, and we report it as a range.

Second price, inserted bids, and the provider rules. On the same blocks, a second-price rule that charges each selected consumer the next density in the order funds 10.88 million provider gas at the drawn declarations, against 11.05 million under pay-as-bid, and 8.97 million at least charges, the same as pay-as-bid. Under it, 94% of selected consumers gain by declaring less, by 30% of the declaration on average.

We also tested inserted consumer bids. Under pay-as-bid the builder may fund any unfunded provider and insert the cheapest consumer bid that completes its funding; a gain is counted net of that bid and of the base fee it burns. No insertion gained in 2,000 sampled blocks of the main mixture under T20, nor in 2,000 blocks under T30 or in the 10% and 33% consumer mixtures. This reports the sample: the residual case of Proposition 4 needs unspent payments close to one provider’s cost, which did not occur. Under the second-price rule a search over bids placed just under each selected consumer, which is not exhaustive, found a gain in 0.35% of blocks.

Neither provider rule binds in these runs: simulated tips are at most 12% of fee caps, and fee caps exceed half the base fee. A stress case with provider tips ten times larger shows what a tighter cap would do. With no cap, tips approach fee caps, the pool pays nearly the whole base fee for a provider, and an inserted consumer bid pays the builder in 81% of blocks, by 3.2% of its revenue on average. The gain comes through the allowance link: consumers admitted by the larger allowance pay part of the added provider’s cost. How much of this the cap removes depends on η : at $\eta = 1/4$ the figures are unchanged, at $\eta = 0.5$ they are 80% and 3.1%, and at $\eta = 0.8$ an inserted bid pays in 3.8% of blocks, by 0.02% of revenue. A cap tight enough to remove this case takes most of the tip from the builder; $\eta = 1/4$ is set for the bound of Proposition 3 and does not address it. The case needs provider tips near fee caps, and what tips competition for funding would produce is left to future work.

T20 and T30 cap provider gas at, respectively, 20% and 30% of the hard block limit. These are model parameters, not proposed Ethereum constants. The allowance bounds provider gas, not total occupancy: realized occupancy extends past $T + \bar{P}_t$ because ordinary gas varies block to block. Against a baseline whose

ninetieth percentile is 58.9%, T20 reaches 78.2%. Occupancy remains bounded above by construction: admission is bounded by $K - H_t^0$, so no block exceeds the hard limit, and node requirements are provisioned against K rather than against realized occupancy. Raising utilization is an objective of TLM regulation and not a side effect. Providers are admitted outside the ordinary controller’s target, so ordinary admission and the ordinary base fee are unchanged, and the added gas is demand the baseline suppressed rather than demand displaced from elsewhere. The effect of higher sustained utilization on node load and state growth is left to future work.

7 Limits and research boundary

Simulation scope. The reported single-slot study uses synthetic traffic to evaluate the provider allowance and base-fee response. Complementary work studies multi-slot TSP mechanisms and feedback-based congestion control; those results are not reported here.

Consumer bidding. The settlement rule is first-price and the simulation draws declarations. An equilibrium of declarations, with unequal realized gas, rebidding, and the allowance link, is left to future work; Table 2 brackets its effect on funded gas.

Order-dependent gas. Density and the allowance use realized gas, and a transaction’s realized gas can depend on what executes before it. The builder must propose a block at which the checks pass; how often reordering changes the selected set remains to be measured.

Deployment. The block-local mechanism of § 3 requires protocol support because it changes valid inclusion and settlement for providers with $m_i < b$, and because a verifiable consumer early-region guarantee requires an ordering rule. This paper specifies the mechanism-level rule; a follow-up will develop an EIP-ready transaction encoding and deployment path.

Future work extends the TLM program to welfare, fairness, strategic behavior, and anti-collusion; feedback-based congestion control; cross-slot TSP mechanisms; and applications to L2 and other L1 chains.

8 Conclusion

We introduce *temporal liquidity*: the economically relevant capacity to move execution across admissible positions, together with the value that movement gains or surrenders. A transaction’s position within a block and its execution delay can change its outcome and value. A temporal liquidity market lets users price that difference. A consumer can authorize payment for an early region and for its relative rank among selected consumers, while a provider accepts execution outside that region. Their crossing turns complementary temporal requirements into a self-funded mechanism rather than a one-sided fee bid, adding a time axis to dynamic fee-market design and congestion control.

We then apply this framework to Ethereum through EIP-1559. The block-local extension gives selected consumers early-region assurance and relative rank among selected consumers within it, and can fund below-floor providers that accept exclusion from that region. Our analysis shows why funded provider gas should be excluded from the ordinary EIP-1559 controller: counting it raises the base fee and displaces ordinary demand. Under TLM regulation, a bounded, consumer-funded provider allowance retains the reference ordinary allocation and raises total utilization in the modeled setting.

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A Payment incidence in the main synthetic mixture

This appendix reports secondary incidence results omitted from the main comparison. Columns Baseline, N, T20, and T30 denote unmodified EIP-1559, naïve metering, and TLM regulation with TLA-funded provider-gas allowances of 20% and 30% of the hard block limit. User entries are total class payments per block, including consumer TLA charges, of which the surplus over funded shortfalls is burned, and provider sender payments; builder entries are revenue. Parentheses show percentage change from Baseline. A dagger marks provider totals whose class composition differs from Baseline.

party	Baseline	N	T20	T30
neutral	630.28m	592.85m (−5.94%)	630.28m (0.00%)	630.28m (0.00%)
consumers	270.47m	301.02m (+11.29%)	324.23m (+19.88%)	324.23m (+19.88%)
providers [†]	5.19m	154.47m [†]	282.31m [†]	293.38m [†]
builder	49.89m	46.71m (−6.37%)	61.78m (+23.84%)	62.34m (+24.96%)

Gas-weighted user payment gives a complementary view:

user class	Baseline	N	T20	T30
neutral	30.18	35.02 (+16.03%)	30.18 (0.00%)	30.18 (0.00%)
consumers	30.19	41.51 (+37.51%)	36.19 (+19.88%)	36.19 (+19.88%)
incl. TLA providers [†]	27.68	26.35 [†]	25.10 [†]	24.99 [†]

[†] Provider totals are not percentage-comparable across cases: baseline provider-class gas contains only transactions whose fee caps clear the base fee, whereas TLA admits additional below-floor providers. Consumer entries divide the sum of the ordinary per-gas charge and the realized lump-sum TLA settlement by realized gas; the temporal component is not a per-gas price (§ 3.1). Neither table is a welfare estimate.

B Consumer-load sensitivity

With 10%, 20%, and 33% offered consumer gas, the realized gas of selected consumers averages 7.21%, 9.77%, and 9.92% of K , respectively. Unselected consumer gas averages 0.17, 3.10, and 8.99 million gas: the allowance $K_C = 0.10K$ is slack in most blocks of the first mixture, active in the second, and heavily oversubscribed in the third, where selected gas stays below K_C only because transactions are indivisible. T20 total utilization is 64.45%, 68.49%, and 69.34%, with funded provider gas of 8.64, 11.06, and 11.57 million gas; T30 gives 64.57%, 69.31%, and 70.92% with 8.70, 11.55, and 12.52 million. The 20% early region is a completion boundary, not consumer-reserved capacity.

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